Exp 3 - Design of IIR Filters Using Impulse Invariance and Bilinear Transformation Methods Using MATLAB

Program:

```
clc;
clear;
close all;

% EXPERIMENT 3
% Design of IIR Filters Using Impulse Invariance
% and Bilinear Transformation Methods

%% STEP 1: INPUT PARAMETERS

Fs  = 1000;     % Sampling frequency in Hz
Fp  = 200;      % Passband edge frequency in Hz
Fst = 300;      % Stopband edge frequency in Hz
Ap  = 1;        % Passband attenuation in dB
As  = 40;       % Stopband attenuation in dB

T = 1/Fs;       % Sampling period

fprintf('========== INPUT PARAMETERS ==========\n');
fprintf('Sampling frequency = %g Hz\n', Fs);
fprintf('Passband edge = %g Hz\n', Fp);
fprintf('Stopband edge = %g Hz\n', Fst);
fprintf('Passband attenuation = %g dB\n', Ap);
fprintf('Stopband attenuation = %g dB\n', As);
fprintf('Sampling period = %g seconds\n', T);

%% STEP 2: DESIGN ANALOG BUTTERWORTH FILTER
% Convert edge frequencies from Hz to rad/s

Wp = 2*pi*Fp;
Ws = 2*pi*Fst;

[N, Wn] = buttord(Wp, Ws, Ap, As, 's');
[ba, aa] = butter(N, Wn, 's');

fprintf('\n========== ANALOG FILTER ==========\n');
fprintf('Filter order = %d\n', N);
fprintf('Cutoff frequency = %.4f rad/s\n', Wn);

disp('Analog numerator coefficients:');
disp(ba);

disp('Analog denominator coefficients:');
disp(aa);

%% STEP 3: IMPULSE INVARIANCE METHOD

[bII, aII] = impinvar(ba, aa, Fs);

fprintf('\n========== IMPULSE INVARIANCE ==========\n');

disp('Numerator coefficients bII:');
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disp(bII);

disp('Denominator coefficients aII:');
disp(aII);

%% STEP 4: BILINEAR TRANSFORMATION WITHOUT PREWARPING

[bBL, aBL] = bilinear(ba, aa, Fs);

fprintf('\n========== BILINEAR WITHOUT PREWARPING ==========\n');

disp('Numerator coefficients bBL:');
disp(bBL);

disp('Denominator coefficients aBL:');
disp(aBL);

%% STEP 5: BILINEAR TRANSFORMATION WITH PREWARPING

Wp_pre = 2*Fs*tan(pi*Fp/Fs);
Ws_pre = 2*Fs*tan(pi*Fst/Fs);

fprintf('\n========== PREWARPING ==========\n');
fprintf('Prewarped passband edge = %.4f rad/s\n', Wp_pre);
fprintf('Prewarped stopband edge = %.4f rad/s\n', Ws_pre);

[Np, Wnp] = buttord(Wp_pre, Ws_pre, Ap, As, 's');
[ba_pre, aa_pre] = butter(Np, Wnp, 's');

[bPW, aPW] = bilinear(ba_pre, aa_pre, Fs);

fprintf('\n========== BILINEAR WITH PREWARPING ==========\n');
fprintf('Filter order = %d\n', Np);
fprintf('Analog cutoff frequency = %.4f rad/s\n', Wnp);

disp('Numerator coefficients bPW:');
disp(bPW);

disp('Denominator coefficients aPW:');
disp(aPW);

%% STEP 6: CALCULATE FREQUENCY RESPONSES

[HII, f] = freqz(bII, aII, 2048, Fs);
[HBL, ~] = freqz(bBL, aBL, 2048, Fs);
[HPW, ~] = freqz(bPW, aPW, 2048, Fs);

% Magnitude in decibels

MII = 20*log10(abs(HII) + eps);
MBL = 20*log10(abs(HBL) + eps);
MPW = 20*log10(abs(HPW) + eps);

% Phase in degrees

PII = unwrap(angle(HII))*180/pi;
PBL = unwrap(angle(HBL))*180/pi;
PPW = unwrap(angle(HPW))*180/pi;
```

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%% STEP 7: ALL GRAPHS IN ONE WINDOW

figure('Name','Experiment 3 - IIR Filter Analysis', ...
       'NumberTitle','off', ...
       'Color','w');

% GRAPH 1: Magnitude response comparison

subplot(2,2,1);

plot(f, MII, 'b', 'LineWidth', 1.4);
hold on;
plot(f, MBL, 'r--', 'LineWidth', 1.4);
plot(f, MPW, 'g-.', 'LineWidth', 1.4);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Magnitude Response Comparison');

legend('Impulse Invariance', ...
       'Bilinear Without Prewarping', ...
       'Bilinear With Prewarping', ...
       'Location','best');

xlim([0 Fs/2]);
ylim([-100 5]);

% GRAPH 2: Phase response comparison

subplot(2,2,2);

plot(f, PII, 'b', 'LineWidth', 1.4);
hold on;
plot(f, PBL, 'r--', 'LineWidth', 1.4);
plot(f, PPW, 'g-.', 'LineWidth', 1.4);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (Degrees)');
title('Phase Response Comparison');

legend('Impulse Invariance', ...
       'Bilinear Without Prewarping', ...
       'Bilinear With Prewarping', ...
       'Location','best');

xlim([0 Fs/2]);

% GRAPH 3: Impulse invariance magnitude response

subplot(2,2,3);

plot(f, MII, 'b', 'LineWidth', 1.4);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Impulse Invariance Response');
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```
xlim([0 Fs/2]);
ylim([-100 5]);

% GRAPH 4: Effect of prewarping

subplot(2,2,4);

plot(f, MBL, 'r--', 'LineWidth', 1.4);
hold on;
plot(f, MPW, 'g-.', 'LineWidth', 1.4);

grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Effect of Prewarping');

legend('Without Prewarping', ...
       'With Prewarping', ...
       'Location','best');

xlim([0 Fs/2]);
ylim([-100 5]);

sgtitle('Experiment 3: IIR Filter Analysis');

%% STEP 8: STABILITY CHECK

polesII = roots(aII);
polesBL = roots(aBL);
polesPW = roots(aPW);

fprintf('\n========== STABILITY CHECK ==========\n');

disp('Impulse invariance poles:');
disp(polesII);

disp('Bilinear without prewarping poles:');
disp(polesBL);

disp('Bilinear with prewarping poles:');
disp(polesPW);

if all(abs(polesII) < 1)
    disp('Impulse invariance filter is stable.');
else
    disp('Impulse invariance filter is unstable.');
end

if all(abs(polesBL) < 1)
    disp('Bilinear without prewarping filter is stable.');
else
    disp('Bilinear without prewarping filter is unstable.');
end

if all(abs(polesPW) < 1)
    disp('Bilinear with prewarping filter is stable.');
else
    disp('Bilinear with prewarping filter is unstable.');
```

```
end

%% STEP 9: FINAL COMPARISON

fprintf('\n========== FINAL COMPARISON ==========\n');
fprintf('Impulse invariance order = %d\n', N);
fprintf('Bilinear prewarped order = %d\n', Np);

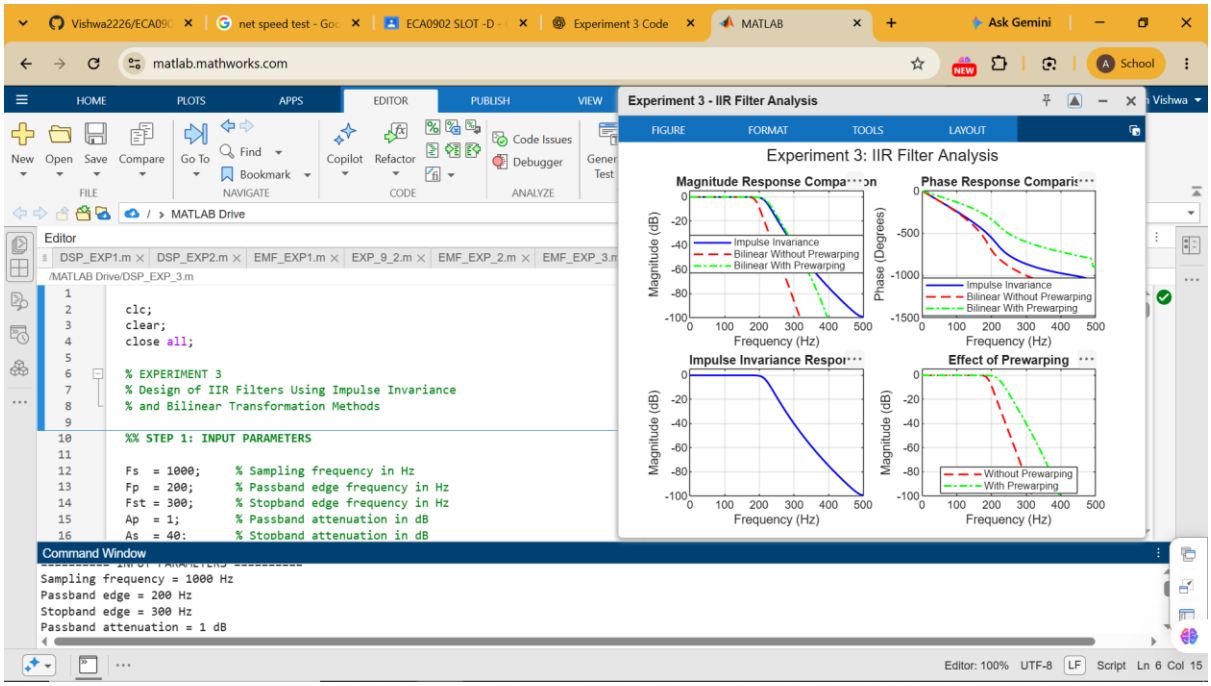
disp('Impulse invariance coefficients:');
disp('bII ='); disp(bII);
disp('aII ='); disp(aII);

disp('Bilinear without prewarping coefficients:');
disp('bBL ='); disp(bBL);
disp('aBL ='); disp(aBL);

disp('Bilinear with prewarping coefficients:');
disp('bPW ='); disp(bPW);
disp('aPW ='); disp(aPW);

disp('Experiment 3 completed successfully.');
```

Output:



Command Prompt :

```
========== INPUT PARAMETERS ==========
Sampling frequency = 1000 Hz
Passband edge = 200 Hz
Stopband edge = 300 Hz
Passband attenuation = 1 dB
Stopband attenuation = 40 dB
Sampling period = 0.001 seconds

========== ANALOG FILTER ==========
```

```
Filter order = 14
Cutoff frequency = 1356.5804 rad/s
Analog numerator coefficients:
   1.0e+43 *

         0         0         0         0         0         0         0         0
0         0         0         0         0         0    7.1489

Analog denominator coefficients:
   1.0e+43 *

    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000    0.0000
0.0000    0.0000    0.0000    0.0000    0.0002    0.0471    7.1489

Warning: The output is not correct/robust. Coeffs of B(s)/A(s) are real, but
B(z)/A(z) has complex coeffs. Probable cause is rooting of high-order repeated
poles in A(s).
B(z) and A(z) have been forced to be real using real(A(z)) and real(B(z)).
> In impinvar (line 104)
In DSP_EXP_3 (line 49)

========== IMPULSE INVARIANCE ==========
Numerator coefficients bII:
    0.0000    0.0000    0.0000    0.0010    0.0117    0.0423    0.0609    0.0387
0.0111    0.0014    0.0001    0.0000    0.0000    0.0000         0

Denominator coefficients aII:
    1.0000   -3.0740    5.8135   -7.6858    7.7240   -6.0974    3.8468   -1.9510
0.7935   -0.2561    0.0643   -0.0121    0.0016   -0.0001    0.0000

Warning: Matrix is close to singular or badly scaled. Results may be inaccurate.
RCOND =  2.549911e-79.
> In bilinear (line 130)
In DSP_EXP_3 (line 61)

Warning: Matrix is close to singular or badly scaled. Results may be inaccurate.
RCOND =  2.549911e-79.
> In bilinear (line 131)
In DSP_EXP_3 (line 61)

Warning: Matrix is close to singular or badly scaled. Results may be inaccurate.
RCOND =  2.516555e-79.
> In bilinear (line 132)
In DSP_EXP_3 (line 61)

Warning: Matrix is close to singular or badly scaled. Results may be inaccurate.
RCOND =  2.516555e-79.
> In bilinear (line 133)
In DSP_EXP_3 (line 61)


========== BILINEAR WITHOUT PREWARPING ==========
Numerator coefficients bBL:
    0.0000    0.0002    0.0012    0.0049    0.0134    0.0268    0.0402    0.0459
```

```
0.0402    0.0268    0.0134    0.0049    0.0012    0.0002    0.0000

Denominator coefficients aBL:
    1.0000   -3.3696    6.9454   -9.7496   10.3578   -8.5415    5.6034   -2.9355
1.2286   -0.4063    0.1042   -0.0200    0.0027   -0.0002    0.0000


========== PREWARPING ==========
Prewarped passband edge = 1453.0851 rad/s
Prewarped stopband edge = 2752.7638 rad/s
Warning: Matrix is close to singular or badly scaled. Results may be inaccurate.
RCOND =  4.155374e-50.
> In bilinear (line 130)
In DSP_EXP_3 (line 83)

Warning: Matrix is close to singular or badly scaled. Results may be inaccurate.
RCOND =  4.155374e-50.
> In bilinear (line 131)
In DSP_EXP_3 (line 83)

Warning: Matrix is close to singular or badly scaled. Results may be inaccurate.
RCOND =  4.126473e-50.
> In bilinear (line 132)
In DSP_EXP_3 (line 83)

Warning: Matrix is close to singular or badly scaled. Results may be inaccurate.
RCOND =  4.126473e-50.
> In bilinear (line 133)
In DSP_EXP_3 (line 83)


========== BILINEAR WITH PREWARPING ==========
Filter order = 9
Analog cutoff frequency = 1650.2477 rad/s
Numerator coefficients bPW:
    0.0021    0.0186    0.0745    0.1739    0.2609    0.2609    0.1739    0.0745
0.0186    0.0021

Denominator coefficients aPW:
    1.0000   -1.0893    1.6925   -1.0804    0.7329   -0.2722    0.0916   -0.0174
0.0024   -0.0001


========== STABILITY CHECK ==========
Impulse invariance poles:
0.1898 + 0.8379i
0.1898 - 0.8379i
0.1829 + 0.6121i
0.1829 - 0.6121i
0.1991 + 0.4432i
0.1991 - 0.4432i
0.2200 + 0.3137i
```

0.2200 - 0.3137i
0.2380 + 0.2095i
0.2380 - 0.2095i
0.2505 + 0.1204i
0.2505 - 0.1204i
0.2568 + 0.0393i
0.2568 - 0.0393i

Bilinear without prewarping poles:
0.3349 + 0.8363i
0.3349 - 0.8363i
0.2830 + 0.6711i
0.2830 - 0.6711i
0.2475 + 0.5265i
0.2475 - 0.5265i
0.2232 + 0.3965i
0.2232 - 0.3965i
0.2070 + 0.2767i
0.2070 - 0.2767i
0.1970 + 0.1635i
0.1970 - 0.1635i
0.1923 + 0.0541i
0.1923 - 0.0541i

Bilinear with prewarping poles:
0.1622 + 0.8261i
0.1622 - 0.8261i
0.1274 + 0.5703i
0.1274 - 0.5703i
0.1084 + 0.3602i
0.1084 - 0.3602i
0.0988 + 0.1747i
0.0988 - 0.1747i
0.0958 + 0.0000i

Impulse invariance filter is stable.
Bilinear without prewarping filter is stable.
Bilinear with prewarping filter is stable.

========== FINAL COMPARISON ==========
Impulse invariance order = 14
Bilinear prewarped order = 9
Impulse invariance coefficients:
bII =
0.0000 0.0000 0.0000 0.0010 0.0117 0.0423 0.0609 0.0387 0.0111 0.0014 0.0001 0.0000
0.0000 0.0000 0

aII =
1.0000 -3.0740 5.8135 -7.6858 7.7240 -6.0974 3.8468 -1.9510 0.7935 -0.2561 0.0643 -
0.0121 0.0016 -0.0001 0.0000

Bilinear without prewarping coefficients:
bBL =
0.0000 0.0002 0.0012 0.0049 0.0134 0.0268 0.0402 0.0459 0.0402 0.0268 0.0134 0.0049 0.0012 0.0002 0.0000

aBL =
1.0000 -3.3696 6.9454 -9.7496 10.3578 -8.5415 5.6034 -2.9355 1.2286 -0.4063 0.1042 -0.0200 0.0027 -0.0002 0.0000

Bilinear with prewarping coefficients:
bPW =
0.0021 0.0186 0.0745 0.1739 0.2609 0.2609 0.1739 0.0745 0.0186 0.0021

aPW =
1.0000 -1.0893 1.6925 -1.0804 0.7329 -0.2722 0.0916 -0.0174 0.0024 -0.0001

Experiment 3 completed successfully.
>>